

EduVision Programming

New Student Assessment Paper

Basic Information

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Current Year: 7

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Past experience learning C++ (terms): No

Past experience learning Python (terms): Yes, around two years.

Part one --- Self-Assessment Trial Questions

These questions are optional. You are encouraged to attempt them. Please submit the source code together with the completed questionnaire.

Problem 1: Numbers Containing Digit 3 or 7

Given two integers m, n , find the number of integers from m to n (inclusive) which contain the digit 3 or 7.

Input Format

One line containing two integers m, n separated by spaces.

Output Format

Output one line containing one integer which is the number of integers from m to n that contain the digit 3 to 7.

Example

Input #1:

13 37

Output #1:

Explanation

Numbers satisfying the requirements are: 13, 17, 23, 27, 30, 31, 32, 33, 34, 35, 36, 37.

Constraints

$$1 \leq m \leq n \leq 10^{18}$$

$$0 \leq n - m \leq 10^7$$

Problem 2: Taxi Fare**Description**

A and B are two places d kilometers apart. Bob takes a taxi from A to B at h hours m minutes. Along the way, there are n checkpoints. Checkpoint i is s_i kms from A, the check takes t_i minutes. Taxi fare rules are:

Normal time (7:00–23:00): \$3 per km, \$2 per waiting minute.

Night time (23:00–7:00): \$5 per km, \$4 per waiting minute.

The taxi drives at a constant speed of 1 km per minute (ignoring acceleration and deceleration). Write a program to calculate how much Bob pays for the trip.

Input Format

1st line: four integers d, h, m, n , followed by n lines, each containing two integers s, t .

Output Format

One line, containing one integer which is the taxi fare.

Example

Input #1:

```
13 12 0 4
3 3
6 7
10 12
12 5
```

Output #1:

Constraints

$$0 \leq h < 24, 0 \leq m < 60$$

$$1 \leq n < d \leq 1000$$

$$1 \leq t_i \leq 100$$

$$\text{For any } 1 \leq i < j \leq n, 1 \leq s_i < s_j < d$$

Problem 3: Sum of All Expressions**Problem Description**

Given a string consisting of digits '1' to '9', you may insert the operator '+' between digits multiple times (or not at all) to form valid expressions. Calculate the sum of the values of all possible valid expressions.

Input Format

One line, containing a string s composed of digits '1' to '9'.

Output Format

One integer, which is the sum of the values of all valid expressions.

Example

Input #1:

125

Output:

176

Explanation #1:

All possible valid expressions are

$$125$$

$$1 + 25 = 26$$

$$12 + 5 = 17$$

$$1 + 2 + 5 = 8$$

Input #2:

9999999999

Output #2:

12656242944

Constraints

$1 \leq |s| \leq 10$

Part two ----- Circle your answer next to each question

Syntax and Standard Library

Input, Output, and Variables:	Not learned	<u>Learned</u>	Proficient
Conditional Branching:	Not learned	<u>Learned</u>	Proficient
Loops:	Not learned	<u>Learned</u>	Proficient
One-Dimensional Arrays:	<u>Not learned</u>	Learned	Proficient
Two-Dimensional Arrays:	<u>Not learned</u>	Learned	Proficient
Character Arrays:	<u>Not learned</u>	Learned	Proficient
Strings:	<u>Not learned</u>	Learned	Proficient
Custom Functions:	<u>Not learned</u>	Learned	Proficient
Struct:	<u>Not learned</u>	Learned	Proficient
Files and Basic I/O:	<u>Not learned</u>	Learned	Proficient
STL Templates:	<u>Not learned</u>	Learned	Proficient

Algorithms and Data Structures (Entry Level)

Enumeration:	<u>Not learned</u>	Learned	Proficient
Simulation:	<u>Not learned</u>	Learned	Proficient
Bubble Sort:	<u>Not learned</u>	Learned	Proficient
Quick Sort:	<u>Not learned</u>	Learned	Proficient
Queues:	<u>Not learned</u>	Learned	Proficient

Stacks:	Not learned Learned Proficient
Linked Lists:	Not learned Learned Proficient
Trees and Binary Trees:	Not learned Learned Proficient
Binary Search:	Not learned Learned Proficient
Binary Lifting	Not learned Learned Proficient
High-Precision Arithmetic	Not learned Learned Proficient
Recursion	Not learned Learned Proficient
Greedy Algorithms	Not learned Learned Proficient
Depth-First Search (DFS)	Not learned Learned Proficient
Breadth-First Search (BFS)	Not learned Learned Proficient
Basics of Dynamic Programming (DP)	Not learned Learned Proficient
1D Dynamic Programming	Not learned Learned Proficient
Knapsack Problems	Not learned Learned Proficient
Interval DP	Not learned Learned Proficient
Euclidean Algorithm	Not learned Learned Proficient
Prime Sieves: Sieve of Eratosthenes	
& Linear Sieve	Not learned Learned Proficient
Sets	Not learned Learned Proficient
Addition Principle (Rule of Sum) &	
Multiplication Principle (Rule of Multiplication)	Not learned Learned Proficient
Permutations and Combinations	Not learned Learned Proficient
STL	Not learned Learned Proficient
Prefix Sum	Not learned Learned Proficient
Difference Array	Not learned Learned Proficient

Algorithms and Data Structures (Advanced Level)

Shortest Path	Not learned	Learned	Proficient
Single-Source Second Shortest Path	Not learned	Learned	Proficient
Minimum Spanning Tree (MST)	Not learned	Learned	Proficient
Second Minimum Spanning Tree	Not learned	Learned	Proficient
Topological Sorting	Not learned	Learned	Proficient
Eulerian Path and Eulerian Circuit	Not learned	Learned	Proficient
Bipartite Graph Determination	Not learned	Learned	Proficient
Strongly Connected Components (SCC)	Not learned	Learned	Proficient
Articulation Points & Bridges	Not learned	Learned	Proficient
Union-Find (Disjoint Set Union)	Not learned	Learned	Proficient
Fenwick Tree (Binary Indexed Tree)	Not learned	Learned	Proficient
Segment Tree	Not learned	Learned	Proficient
Splay Tree	Not learned	Learned	Proficient
Treap	Not learned	Learned	Proficient
Center, Diameter and Centroid of a Tree	Not learned	Learned	Proficient
DFS Order and Euler Tour of a Tree	Not learned	Learned	Proficient
Tree Difference, Subtree Sum & Doubling	Not learned	Learned	Proficient
Lowest Common Ancestor (LCA)	Not learned	Learned	Proficient
Discretisation	Not learned	Learned	Proficient
Divide and Conquer	Not learned	Learned	Proficient
Radix Sort	Not learned	Learned	Proficient
Tree DP	Not learned	Learned	Proficient
State Compression DP	Not learned	Learned	Proficient
Common DP Optimisations	Not learned	Learned	Proficient

Euler's Theorem & Euler's Totient Function

~~Not learned~~ Learned Proficient

Bézout's Identity

~~Not learned~~ Learned Proficient

Modular Multiplicative Inverse

~~Not learned~~ Learned Proficient

Extended Euclidean Algorithm

~~Not learned~~ Learned Proficient

Chinese Remainder Theorem

~~Not learned~~ Learned Proficient

Permutations & Combinations on Multisets

~~Not learned~~ Learned Proficient

Derangements & Circular Permutations

~~Not learned~~ Learned Proficient

Inclusion-Exclusion Principle

~~Not learned~~ Learned Proficient

Catalan Numbers

~~Not learned~~ Learned Proficient

Vectors and Matrices

~~Not learned~~ Learned Proficient

Gaussian Elimination

~~Not learned~~ Learned Proficient

String Matching (KMP)

~~Not learned~~ Learned Proficient

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